

1. Ask the user to enter any number. Check whether it is an int or a float using `isinstance()`. Print a message that states the number and its type, e.g. "42 is an integer." or "3.14 is a float."

SAMPLE INPUT

Enter a number: 7.5

SAMPLE OUTPUT

7.5 is a float.

2. Take two numbers from the user as input (they will be strings by default). Convert them to floats, then print their sum, difference, product, and quotient – each rounded to 2 decimal places.

SAMPLE INPUT

First number: 10

Second number: 3

SAMPLE OUTPUT

Sum: 13.0

Difference: 7.0

Product: 30.0

Quotient: 3.33

3. Ask the user to enter a sentence. Then print the following on separate lines: the sentence in uppercase, in lowercase, the total number of characters (including spaces), and the sentence with all spaces replaced by hyphens.

SAMPLE INPUT

Enter a sentence:

Hello World

SAMPLE OUTPUT

HELLO WORLD

hello world

Characters: 11

Hello-World

4. Take a word as input from the user. Print its first character, its last character, and the word reversed – using only slicing and indexing (no built-in `reverse()`).

SAMPLE INPUT

Enter a word: Python

SAMPLE OUTPUT

First: P

Last: n

Reversed: nohtyP

5. Ask the user to enter a sentence and then a word to search for. Using the `in` keyword, check whether the word exists in the sentence. Print a

formatted message announcing the result. The check should be case-insensitive.

SAMPLE INPUT

Sentence: I love Python
Search word: python

SAMPLE OUTPUT

"python" was found in the sentence.

6. Ask the user to enter their age. Based on the value, print one of the following categories: Child (0–12), Teenager (13–17), Adult (18–64), or Senior (65 and above). If the age is negative, print "Invalid age."

SAMPLE INPUT

Enter your age: 16

SAMPLE OUTPUT

Category: Teenager

7. A theme park ride requires riders to be at least 120 cm tall and at most 200 cm tall. Ask the user for their height in cm. Print whether they are "Eligible to ride" or "Not eligible to ride", along with a brief reason if not eligible.

SAMPLE INPUT

Enter your height (cm): 110

SAMPLE OUTPUT

Not eligible to ride.
Reason: Too short (minimum 120 cm).

8. Build a simple bill-splitting calculator. Ask the user for the total bill amount, a tip percentage, and the number of people. Calculate each person's share (bill + tip, divided equally). Use round() to round to 2 decimal places and print the final amount per person.

SAMPLE INPUT

Total bill: 1500
Tip (%): 10
Number of people: 4

SAMPLE OUTPUT

Each person pays: 412.5 BDT

9. Ask the user to enter a username. Apply the following validation rules and print the appropriate message for each violation. If all rules pass, print "Username accepted." Rules: must be at least 5 characters long • must not contain spaces • must start with a letter (not a digit).

SAMPLE INPUT

Enter a username: 9abc

SAMPLE OUTPUT

Error: Username must start with a letter.
Error: Username must be at least 5
characters.

10. Build a basic student grade report. Ask the user for their name and three subject marks (out of 100). Calculate the average using `round()`. Then determine the grade using this scale: A (90–100), B (75–89), C (60–74), D (45–59), F (below 45). Print a formatted report showing the name, each mark, the average, and the final grade.

SAMPLE INPUT

Name: Arif
Math: 88
English: 74
Science: 91

SAMPLE OUTPUT

--- Grade Report ---
Student: Arif
Math: 88 | English: 74 | Science: 91
Average: 84.33
Grade: B